

Casey Baker, PhD

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Machine Learning Engineer | Security Clearance: Top Secret/SCI

EXPERIENCE

Expedition Technology

Herndon, VA

Machine Learning Engineer, Scrum Master

2024 – Present

- Designed an active learning metric for imbalanced 3D point cloud data streams, improving object detection accuracy by 25% on rare classes, maintaining performance on common classes, and reducing manual labeling by 10%
- Developed a pipeline to reduce speckle noise from commercial radar images and train a transformer-based masked autoencoder for object detection in radar images with varying resolutions
- Collaborated in teams of 4–7 developers to design and evaluate deep learning pipelines for trajectory data and remote sensing image analysis, including anomaly detection, adversarial robustness, self-supervised pretraining, and 2D/3D object detection

Duke University

Durham, NC

PhD Candidate advised by Dr. Yiyang Gong, NSF Graduate Research Fellow

2020 – 2023

- Built a deep learning pipeline to predict underlying neural activity using spatiotemporal information from fluorescence videos that improved detection of low signal-to-noise events and improved spike rate prediction by 14%
- Designed a semi-supervised deep learning pipeline for neuron segmentation that used $\frac{1}{4}$ as many ground truth labels as fully supervised methods while maintaining accuracy on par with human labeling
- Developed a computational model of the visual cortex and applied machine learning to identify network properties of optimal pattern completion neurons

Massachusetts Institute of Technology

Cambridge, MA

Senior Research Support Associate advised by Dr. Steven Flavell

2018 – 2020

- Performed and analyzed experiments testing the behavioral effects of activating and silencing neurons in *C.elegans*

PUBLICATIONS

- **Baker, C.**, Daniel, B., Immel, E., Bogart C. (2025). Object Detection on Streaming LiDAR Data with Active Learning. *Military Sensing Symposia Active E-O Systems*
- **Baker C.**, Gong Y. (2023). A semi-supervised pipeline for accurate neuron segmentation with fewer ground truth labels. *eNeuro*.
- **Baker C.**, Gong Y. (2023). Identifying properties of pattern completion neurons in a computational model of the visual cortex. *PLoS Computational Biology*.
- Huang Y-C., Luo J., Huang W., **Baker C.**, Gomes M., Byrne A., Flavell S. (2023). A single neuron in *C. elegans* orchestrates multiple motor outputs through parallel modes of transmission. *Current Biology*.
- Beltzer, M., Moulder, R., **Baker, C.**, Comer, K., & Teachman, B. (2022). Effects of mass shootings on mental illness stigma in the United States. *Personality and Social Psychology Bulletin*.
- Ji, N., Madan, G., Fabre, G., Dayan, A., **Baker, C.**, Kramer, T., Nwabudike, I., Flavell, S. (2021). A neural circuit for flexible control of persistent behavioral states. *Elife*.

EDUCATION

Duke University

Durham, NC

PhD. Biomedical Engineering, Certificate in College Teaching, GPA: 4.0

2020 – 2023

University of Virginia

Charlottesville, VA

B.S. Biomedical Engineering, B.A. Psychology, GPA: 3.91

2014 – 2018

SKILLS

Languages: Python, MATLAB, R

Frameworks/Libraries: Tensorflow, Pytorch, Scikit-learn, OpenCV, NetworkX, NumPy, Pandas, Ray

MLOps: CI/CD, Docker, Git, MLflow, Airflow